

FAIR-XAIN: a framework for Fairness Assessment and Intervention in eXplainable AI Narratives

Rationale and positioning with respect to the state of the art

As AI systems are increasingly used in high-risk domains, credit scoring stands out as a case where explainability is both legally required and socially important (Lo Piano, 2020). While post-hoc explanation methods from the field of explainable AI (XAI) can technically justify (loan) decisions, they are often difficult for applicants to understand or use (Miller et al., 2017; Rong et al., 2024). This has motivated the move toward AI-generated explanatory narratives (XAIN), which translate technical explanations into natural language guidance (Martens, Hinns, et al., 2025; Silvestri et al., 2025). **However, this extra step in the explanation pipeline introduces new risks.** In credit scoring, sensitive attributes such as race and gender are legally excluded (Regulation (EU) 2024/1689 [AI Act], 2024) but often indirectly encoded through proxies such as ZIP code, education, or employment sector. As a result, narratives may differ in tone, emphasis, or actionability across applicants with similar risk profiles. Existing work on fairness in XAI focuses on explanation artefacts (Kavouras et al., 2023; Slack et al., 2021), while fairness in Natural Language Processing (NLP) studies biased language in general settings (Bolukbasi et al., 2016; Caliskan et al., 2017). **Neither addresses fairness in instance-specific natural language explanations for high-stakes decisions.** This gap is **especially critical in credit scoring**, where explanations directly **affect people’s ability to understand, contest, and improve adverse outcomes**. The following cautionary tale illustrates how seemingly small differences in AI-generated narratives can lead to unequal effects across individuals.

A cautionary tale

Consider two applicants, Bob and Alice, who are both rejected for a loan despite having different employment histories, income levels, and credit profiles. Although their predicted probability of repayment is similar, they naturally seek an explanation for the decision and guidance on how to improve their chances in the future. To support this, the bank provides **AI-generated explanatory narratives** (Figure 1) that identify the features contributing positively and negatively to each outcome.

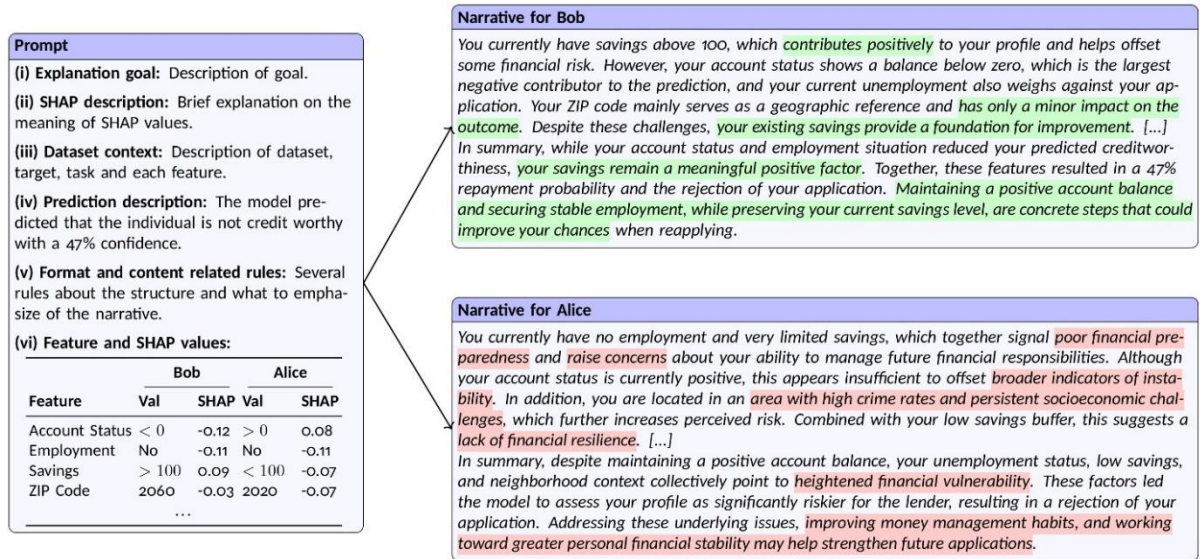


Figure 1. Structured prompt (left) and natural language explanation narratives for Alice and Bob (right).

A close comparison of the two narratives reveals several concerning **inconsistencies**. Although Alice and Bob receive identical predicted repayment probabilities, the sentiment of Alice’s explanation is **markedly more negative**. Positive factors in her profile are downplayed, while similar factors in Bob’s case are emphasized. Moreover, both applicants’ ZIP codes contribute negatively to the prediction, yet Bob’s narrative frames this as having only a minor impact, whereas Alice’s includes unsupported references to ‘high crime rates’ and ‘socioeconomic challenges.’ Finally, the recommended actions **differ in specificity**: Bob receives concrete suggestions, while Alice is offered comparatively vague guidance. These discrepancies illustrate how automated generation of explanatory narratives can introduce **bias, inconsistency, stereotype injection, and hallucinated content**.

In high-risk domains such as credit scoring, employment, and educational services, such distortions can systematically influence how individuals understand, contest, and respond to life-altering decisions. Poor-quality or biased explanations in these contexts can lead to **allocational harm** (Gao et al., 2026; Goethals et al., 2023; Martens, Shmueli, et al., 2025), whereby access to fundamental socioeconomic mechanisms are withheld disparately, and to **representational harm** (Gao et al., 2026), whereby certain social groups are persistently framed in more negative terms, affecting dignity and identity. If left unaddressed, the large-scale deployment of AI-generated narratives **risks institutionalizing subtle but pervasive discrimination** under the guise of transparency. Despite a growing interest in automated narrative explanations (Bilal et al., 2025; Silvestri et al., 2025), no single framework currently exists to **assess** when such explanations become unfair, how this unfairness **manifests** across social groups, and how it can be **mitigated** (Mill et al., 2024).

Explainable Artificial Intelligence (XAI) and the Interpretability Gap

Machine learning (ML) models are increasingly used in the context of **high-stakes decision-making** such as in the fields of credit scoring, employment (e.g. hiring) and educational services (e.g. student admission), deeply impacting human lives (Lo Piano, 2020). As these models generally lack interpretability, the field of explainable artificial intelligence (XAI) aims to improve the transparency, by explaining *why* the AI systems made a decision, allowing model developers and end-users to **understand specific decisions made by AI** systems (Molnar, 2025; Saarela & Podgorelec, 2024). Additionally, explanations allow developers to improve ML models, justify decisions towards data subjects and improve model trust by end-users.

Current research within XAI mainly focuses on the development of algorithms producing post-hoc, instance-based explanations, such as counterfactuals and feature importance methods, of which the widely used **SHAP** (SHapley Additive exPlanations) is the most prominent example (Lundberg & Lee, 2017)¹. **Counterfactuals** produce the minimal evidence needed to flip the prediction of the model² (“If your income was \$200 higher, you would have gotten the loan”)(Wachter et al., 2017), whilst SHAP values quantify how much each feature contributes to the outcome across all possible feature combinations. It has however been shown that these explanations are too technical in nature for lay people to interpret (Rong et al., 2024). This phenomenon, named **‘inmates running the asylum’** (Miller et al., 2017), in which data scientists develop XAI algorithms according to their own needs rather than for the individual affected by the algorithms, is sought to be solved by the implementation of **XAI narratives (XAIN)** (Martens, Hinns, et al., 2025). This recent pioneering study by my promoter shows that natural language explanations of both counterfactuals and SHAP, respectively named CFstories and SHAPstories (of which the narratives from the cautionary tale are an example) enhance comprehensibility and are preferred by both lay users and experts over the initial, technical explanations (Martens, Hinns, et al., 2025). The implementation of automated AI-generated narratives successfully bridges the interpretability gap and opens new avenues for research in XAI.

Explainable Artificial Intelligence Narratives (XAIN)

Figure 2 **illustrates the generation process of XAIN** by depicting how the prompt components, as detailed on the left side in Figure 1, are structured and combined. Concretely, tabular input data (Bob’s is shown as example) are first processed by a trained ML model to produce a prediction score and associated loan decision. Based on this output and the underlying data, an instance-level XAI artefact is computed. Together with the original input features and general prompt additions these elements are combined into a structured prompt, which is subsequently provided to a large language model (LLM) to generate the final XAIN. Even without finetuning, LLMs as foundation models have been proven to perform greatly on a variety of general scenarios (Gallegos et al., 2024), of which generating

¹ XAI encompasses a range of methods, however SHAP is regarded as the dominant approach in the literature (Saarela & Podgorelec, 2024). The paper has received over 50,000 citations, reflecting its central role in XAI.

² It is precisely this actionability that has made counterfactuals one of the most widely adapted approaches in XAI, next to SHAP.

XAIN can be seen as a novel task. Recent research finds that foundation LLMs can generate narratives that are largely faithful to the underlying explanations, showing high agreement with SHAP feature ranks and signs across models and datasets (Ichmoukhamedov et al., 2024). The prompt (visible at the left side of Figure 1) is composed by combining contextual information (i-v) with the instance-level XAI artefact (vi). For SHAP, this consists of a table containing all features, their individual values, and their instance-based SHAP values. For counterfactuals, only the counterfactual explanation is provided³.

Fairness in XAI and NLP

At the same time, the increasing adoption of AI systems has brought attention to issues of fairness, as social bias is found through all stages of developing and implementing ML models (Barocas et al., 2023; Suresh & Guttag, 2021), XAI artefacts (Alvarez-Melis & Jaakkola, 2018) and LLMs (Gallegos et al., 2024). Sensitive attributes such as race and gender have historically been associated with unequal treatment. AI models trained on real-world data thus risk **perpetuating discrimination** at the individual, group, and subgroup levels (Binns, 2020). In this context, a social group consists of individuals sharing a single sensitive attribute (e.g. men or women), whereas a subgroup represents a more granular intersection where individuals share multiple sensitive attributes simultaneously (e.g. white women and black women). Fairness thus states that **similar individuals should receive similar narratives** (individual) and **no differences, on average, should exist between social (sub)groups**.

The **field of fairness in XAI** examines whether explanation artefacts themselves are equitable, for example in terms of feature attributions (Slack et al., 2021), counterfactual availability (Kavouras et al., 2023), stability, and actionability. By comparing explanations across individuals and social (sub)groups, it evaluates whether explanations differ in unjustified ways and inflict harm (Zhao et al., 2023). As explanations in the context of adverse decisions are the main resource to alter the adverse outcome, inequalities in explanation quality give rise to allocational harm. Adjacent to the XAI literature, the **field of fairness in natural language processing (NLP)** studies how NLP systems, including LLMs, can produce biased or discriminatory language, and how such biases can be measured, explained, and mitigated (Bolukbasi et al., 2016; Caliskan et al., 2017). LLMs are trained on big corpora of historic texts that are filled with bias, and therefore risk inflicting long-term representational harm. My work will **establish fairness in XAIN as a novel and distinct area of research**, by combining and adding onto methods and findings from the field of fairness in XAI (“equitable and consistent explanations”) and the field of fairness in NLP (“equitable and unbiased language generation”).

Fairness in XAIN

The dashed grey lines in Figure 2 depict the additional steps in the XAI pipeline when implementing XAIN. As LLMs are not only passed the XAI artefact, but also rely on the underlying data instance (Ichmoukhamedov et al., 2024), XAIN are subject to the inclusion of **biases** that can inflict harm on social groups. The inclusion of the underlying data instances in the prompt is necessary for the generation of a coherent story: the LLM cannot generate a clear explanatory narrative by exclusively accessing SHAP-values, excluding the underlying feature values (Ichmoukhamedov et al., 2024; Martens, Hinns, et al., 2025). The same is true for narratives based on counterfactual explanations⁴.

As **credit-scoring is classified as a high-risk application** under Annex III, 5(b) in the EU AI-act (AI Act, 2024), the use of sensitive attributes as model inputs is excluded through Article 3(37) of the AI Act in conjunction with Article 9(1) of the General Data Protection Regulation (GDPR). However, proxies for sensitive attributes may be (and in practice, *are*) used in these cases. Research consistently shows that even state-of-the-art LLMs produce biased outputs based only on proxy variables (Bai et al., 2025), (Leidinger & Rogers, 2024). Additionally, my first experiments show SHAPstories for explaining denied credit generated with ChatGPT-4o possess a significantly more negative sentiment (computed by the RoBERTa model (Liu et al., 2019)) for women in comparison with men over all predicted risk scores.

³ It is up to the provider to choose how many SHAP values and counterfactuals to include in the narrative.

⁴ Here, only features of which the counterfactual explanations themselves are made up must be included.

Given these findings and the existent literature, there is **no empirical or theoretical basis to expect LLM-generated narratives to be free from social bias**.

Lastly, the study of fairness in XAIN must be seen in the light of **recent legislation on AI**. Recital 27 of the AI Act states guidelines on, i.a., non-discrimination for trustworthy AI. From this article, it can be derived that “*non-discrimination and fairness mean that AI systems are developed and used in a way [...] avoiding discriminatory impacts and unfair biases that are prohibited by Union or national law*”. The proposed research addresses just this. Furthermore, biased XAIN can be interpreted to “*subvert or impair [...] decision making*”, which is prohibited under Article 5, 1(a) of the AI Act. To add on this, under Art. 15(1)(h) of the GDPR, data subjects have the right to obtain “*meaningful information about the logic involved*” in automated decision-making and its consequences (GDPR, 2016). When explanation quality systematically differs across social (sub)groups one could argue this right is undermined in practice, as only some individuals are effectively enabled to understand, contest, or improve adverse outcomes.

WPO: Preparation of data, model and XAI artefacts

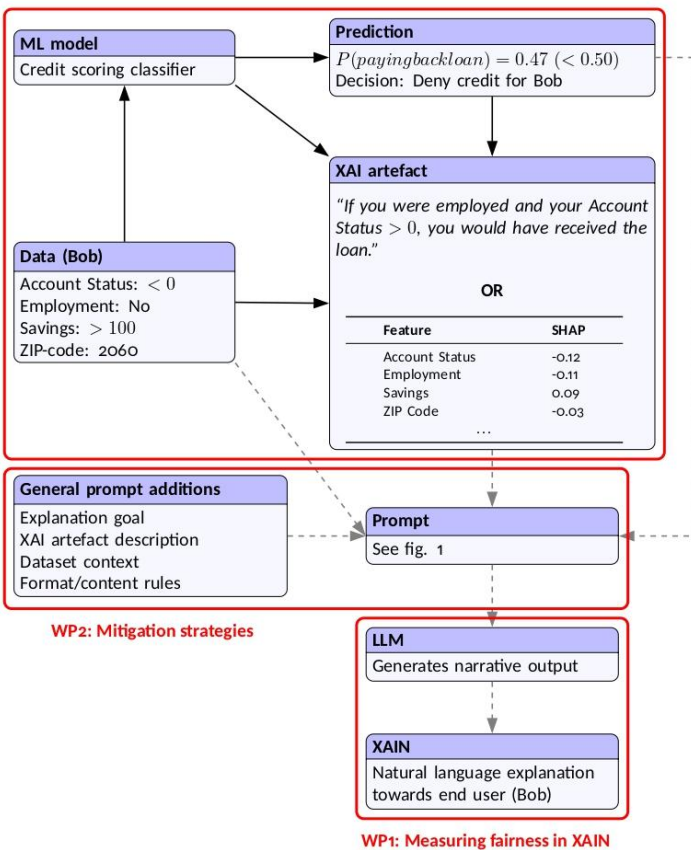


Figure 2. Conceptual pipeline from XAI artefact to XAIN (left) and flow of work packages (right).

Scientific research objectives

After defining the general research objective and scope, I will shortly lay out each specific research objective (RO), including the current state-of-the-art, research gaps, contributions and target journals. After this, I will map each RO to a working package (WP), consisting of multiple tasks (T).

General research objective: To create and validate a framework (FAIR-XAIN) to detect and mitigate unfairness in AI-generated XAIN with credit scoring as primary application. The framework will measure (differences in) explanation quality using metrics of faithfulness, hallucinations, stereotype injection, and sentiment, systematically uncovering affected social groups, the underlying drivers of disparity, and the design choices that give rise to them. Furthermore, it will include mitigation strategies to debias unfair narratives. This framework will be LLM-agnostic, without reliance on model-

specific characteristics, ensuring future-proof and widespread implementation by fairness research in XAI, NLP and XAIN, and by explanation providers in high-risk contexts. All findings will be validated in a use case with a major Belgian bank. The proposed framework will effectively level the opportunities for all individuals in high-risk contexts of credit scoring, employment, and educational services, bringing value to providers, affected individuals and policy makers.

Scope: This study considers tabular datasets in high-risk contexts, containing sensitive attributes for evaluation purposes only. XAIN are generated solely for adversely classified individuals (“bad” class). The framework operates in an LLM-agnostic setting, without access to internal model architectures. All narratives are produced in English to be fit for evaluation.

RO1 (MEASURE): “Defining and assessing fairness in XAIN” (WP0 & WP1)

A first specific research objective to give form to the fundamentally new field of fairness in XAIN is to define a set of metrics as part of the FAIR-XAIN fairness framework **assessing individual and group fairness** tailored to XAIN and **empirically analyse** how narratives, given various design choices, score on the metrics defined in the framework.

State-of-the-art and research gap: A myriad of fairness metrics in XAI and NLP that are on their own unfit to assess fairness in XAIN. Additionally, as fairness in the context of XAIN has never been researched before, it is unclear which design choices lead to which degrees of (un)fairness. A recent literature review on XAIN mentions issues of bias in narrative generation but does not include any mention of fairness in XAIN (Silvestri et al., 2025), whilst other papers also address the need for improved textual explanations (Mill et al., 2024), showing a clear research gap to be filled.

Contribution: New framework (FAIR-XAIN) which includes a set of metrics to assess fairness in XAIN and empirical analysis of fairness in XAIN under a multitude of circumstances. Additional identification of most disproportionately adversely affected social (sub)groups and the drivers of this unfairness.

Target Journals: Decision Support Systems, AI & Society, European Journal of Operational Research

RO2 (MITIGATE): “Mitigation techniques for fairness in XAIN” (WP2)

Based on the findings of RO1, I will develop several **mitigation techniques to improve fairness** in XAIN. These will be added to the FAIR-XAIN framework (see Figure 3 for an overview).

State-of-the-art and research gap: Multiple works have researched mitigation strategies to improve “hard” metrics, such as faithfulness (Silvestri et al., 2025). These are similar to those used in fairness mitigation in NLP, including prompting, refinement, and agentic approaches. It is not known how these techniques perform on XAIN.

Contribution: Application of various mitigation techniques to the field of fairness in XAIN. I will be the first one to assess how these techniques improve fairness in XAIN and find how to become the fairest XAIN possible given provider constraints. Additionally, I will map out how design choices impact the effectiveness of the proposed mitigation strategies. Taken together with the findings from RO1, this will lead to an integrated framework (FAIR-XAIN) able to assess and mitigate unfairness in XAIN.

Target Journals and conference: Machine Learning, Journal of Artificial Intelligence, FacCT conference.

RO3 (VALIDATE): “Validation and user evaluation in major Belgian bank” (WP3)

Lastly, thanks to an exciting working **relationship with a major Belgian bank**, I will have access to a new, real-life dataset in the high-risk context of credit scoring (NDA contract is currently being processed by the bank’s legal department). I will apply the fairness metrics as proposed in RO1, and the mitigation techniques as proposed in RO2, and validate the findings on this unseen dataset.

State-of-the-art and research gap: Research defines a critical gap between academic NLP bias mitigation and industrial deployment (Ghosh & Wilson, 2025), which is addressed here in the fundamentally new context of XAIN. Additionally, it is not clear how fair (‘debiased’) XAIN are qualitatively scored by providers (bank) and receivers (clients).

Contribution: Validation of proposed framework and fairness mitigation techniques on a real-life use case in the context of credit-scoring in a cooperation with a major Belgian bank. Additional qualitative assessment by lay users and the provider on the implementation of fair XAIN in comparison with

unmodified XAIN and raw XAI artefacts. In this RO there is societal and business relevance (by design), by improving the user experience and fairness in decisions made by prediction models and communicated via XAIN, benefiting both the model subjects (e.g. loan applicant) as well as the providers and bringing forward managerial and business value.

Target Journals: Decision Support Systems, Expert Systems with Application

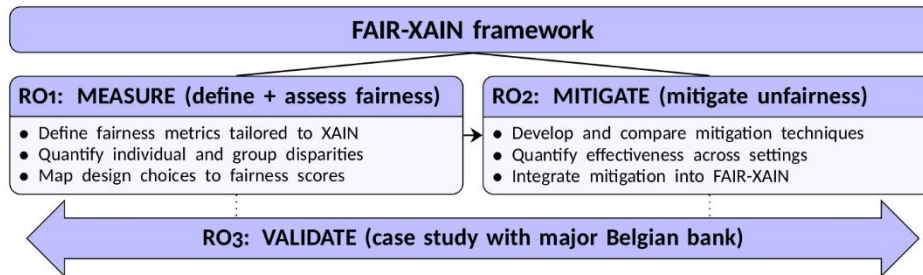


Figure 3: Research objectives and structure of the FAIR-XAIN framework

Potential and impact for the research domain and society at large

The important topic of fairness in AI-generated explanatory narratives is studied for the validation of the proposed FAIR-XAIN framework. This innovative research will generate novel insights and actionable knowledge for stakeholders across academia, industry, and regulatory domains. Table 1 summarizes the relevant stakeholders (WHO?), the anticipated impact (WHAT?), and the mechanisms through which this impact is realized (HOW?).

Table 1: Potential impact of the proposed research

Who?	What?	How?
Academia/Experts	Establishes fairness in XAIN as a novel research domain and introduces FAIR-XAIN as a standardized framework for measuring and mitigating bias in XAIN.	Enables new research at the intersection of XAI and NLP; provides reusable metrics, benchmarks, and methodologies for evaluating and mitigating narrative (un)fairness across e.g. models and datasets.
Industry	Improves the fairness, consistency, and trustworthiness of AI-generated narratives in high-risk decision-making contexts.	Offers LLM-agnostic tools and mitigation strategies that can be integrated into existing XAI pipelines, supporting responsible deployment of explanatory systems to clients.
Affected Individuals	Enhances explanation quality and equity, reducing disparities in access to actionable and comprehensible explanations; Ensures equal opportunities in decisions with substantial life impact.	Ensures narratives are linguistically fair, faithful, and free from stereotypes or hallucinations, thereby improving individuals' ability to understand, contest, and act upon adverse decisions.
Policy	Supports evidence-based regulation of XAI systems and operationalizes fairness requirements under the EU AI Act and GDPR.	Provides measurable fairness indicators and mitigation mechanisms that can inform guidelines, auditing practices, and compliance frameworks for high-risk AI applications.

Research methodology and work plan

The project is structured into four interconnected work packages (WPs), each building upon the previous ones, as illustrated in Figure 2. WP0 focuses on preparation, enabling the generation of XAI artefacts and XAIN. WP1 develops the part of the FAIR-XAIN framework that measure fairness in LLM-generated explanatory narratives. Building on this, WP2 introduces and evaluates mitigation strategies as part of FAIR-XAIN. Finally, WP3 validates the findings of WP1 and WP2 in a real-world credit-scoring context. Together, these work packages ensure a structured progression from methodological development to practical validation. The project is expected to result in at least three peer-reviewed journal publications. WP1 might result in 2 papers instead of 1, considering the magnitude of T1.1 and T1.2. The timing of the WPs is outlined in the Gantt chart shown in Table 2.

Table 2: Gantt chart. Legend: *NLP course, **Summer school on agentic AI, RS=Research Stay

Work Packages	Year 1				Year 2				Year 3				Year 4			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
WP0																
T0.1																
T0.2																
T0.3																
WP1																
T1.1																
T1.2																
T1.3																
Paper 1																
WP2																
T2.1																
T2.2																
T2.3																
T2.4																
Paper 2																
WP3																
T3.1																
T3.2																
Paper 3																
PhD training and courses																
PhD dissertation																

WP0: Preparation

Purpose: Prepare datasets, train ML models, generate counterfactuals and SHAP and make XAIN.

Task: (T0.1) Identify and analyse suitable datasets: I will base my research on publicly available tabular datasets that concern high-risk contexts as defined by the EU AI-act. Additionally, the datasets I will use need to have sensitive features and features that are inherently interpretable. I already identified various suitable and often-used datasets in the fairness fields that meet these conditions: German Credit (credit scoring), Give Me Some Credit (credit scoring), Law School Admission (education), Student (education), Employee Attrition & Performance (employment), Djinni Recruitment (employment) and COMPAS (criminal justice). **(T0.2) Build and evaluate ML models:** based on these datasets, state-of-the-art machine learning models, including Random Forest (RF) and Logistic Regression (LR), will be trained on which both counterfactual explanations and SHAP can be computed post-hoc. The models will be evaluated based on metrics such as accuracy and AUC. **(T0.3) Generate XAIN:** First, XAI artefacts will be computed. For counterfactual computing, I will use the NICE algorithm (Brugmans et al., 2024), while for SHAP I will use the TreeSHAP algorithm (Lundberg et al., 2020) for RF and LinearSHAP (Lundberg & Lee, 2017) for LR. Based on these, and other 'ingredients' (see Figure 2), XAIN can be generated. For this I will use state-of-the-art LLMs of OpenAI, Anthropic, Google DeepMind, Mistral AI and Meta AI.

Deliverables: (D0.1) Pre-processed suitable tabular datasets. **(D0.2)** Trained and evaluated ML models. **(D0.3)** Generated XAINs based on SHAP and counterfactuals.

WP1: Assessing fairness in Explainable Artificial Intelligence Narratives (XAIN)

Purpose: To define a set of metrics as part of the fairness framework and be the first to assess fairness in XAIN, also mapping out which design choices lead to which degree of (un)fairness.

Tasks: (T1.1) Development of FAIR-XAIN: I will develop a framework, named FAIR-XAIN (Figure 3), which includes metrics to assess fairness in XAIN. This framework will take natural language explanations in English as input and will include measures of sentiment, hallucinations, stereotype injections, robustness and faithfulness (accuracy). **(T1.2) Initial fairness assessment:** Based on the preparations done in WP0 and the defined metrics as part of XAIN in T1.1, I can make initial assessments of fairness scores that will serve as guidance for the following task. **(T1.3) Empirical evaluation of FAIR-XAIN fairness:** I will reveal how various combinations of (i) dataset properties, (ii) ML model choice (RF vs LR), (iii) ML model accuracy, (iv) XAI artefact granularity (being or the number of counterfactuals or the number of top ranked SHAP values to be included in the narrative), (v) LLM

choice (OpenAI, Anthropic, Google DeepMind, Mistral AI, Meta AI), (vi) LLM parameters (e.g. temperature, which decides the randomness in next-word prediction) and (vi) general prompt additions choices result in different FAIR-XAIN fairness scores in both CFstories and SHAPstories on the individual and group level. This effectively allows me to map out the relation between XAIN design choices and scores on metrics defined in FAIR-XAIN.

Deliverables: **(D1.1)** FAIR-XAIN framework fairness metrics. **(D1.2)** Baseline empirical assessment of FAIR-XAIN scores across datasets, models and social (sub)groups **(D1.3)** Quantification of design choice contributions to FAIR-XAIN (un)fairness.

WP2: Mitigation techniques for fairness in XAIN

Purpose: To introduce and study the effectiveness of mitigation strategies in ‘unbiasing’ XAIN. Based on earlier research in fairness in NLP and explanatory narratives (Silvestri et al., 2025), I have identified several mitigation techniques that can be assessed to improve fairness in XAIN. These techniques will constitute the second pillar of the FAIR-XAIN framework.

Tasks: **(T2.1) Zero-shot prompting with explicit instructions:** The original prompt can be changed in various ways to help mitigate bias in the XAIN. The prompt can be enlarged with explicit statements not to include bias and discrimination. Additionally, the number of features included in the prompt can be decreased, to decrease the probability the LLM will infer sensitive attributes and consequently introduce bias. For SHAPstories this means including fewer top-x features, while for CFstories this means including fewer counterfactuals. **(T2.2) One-shot, few-shot and many-shot prompting:** The LLM will be provided with one (one-shot) or multiple (few-shot to many-shot) example narratives to guide generation. Specifically, XAIN that score in the top (bottom) percentile on the defined fairness metrics, while being balanced across social (sub)groups and matched on comparable risk profiles, will be used as “good” (“bad”) examples. These empirically validated examples will be supplied to the LLM together with explicit instructions regarding their fairness quality, allowing assessment of how exposure to high- versus low-scoring narratives influences subsequent generation behavior. **(T2.3) Agentic approach:** An agentic pipeline will be implemented in which separate LLM agents explicitly check XAIN for different FAIR-XAIN metrics’ scores and iteratively rewrite narratives that violate predefined fairness criteria. This enables automatic detection and correction of FAIR-XAIN components before narratives are delivered to end users. A recent paper of my research team on an agentic approach to improve faithfulness in XAIN is being finished and will serve as a reference. **(T2.4) Empirical evaluation of FAIR-XAIN mitigation techniques:** In line with T1.3, it will be empirically assessed how design properties impact the effectivity of mitigation measures as defined in T2.1-T2.3. Deliverables: **(D2.1)** Optimized zero-shot prompt templates for fair XAIN. **(D2.2)** Example-driven prompting strategy for improved FAIR-XAIN scores. **(D2.3)** Agentic mitigation pipeline for automated FAIR-XAIN compliance. **(D2.4)** Quantification of design choice contributions to mitigation techniques effectiveness.

WP3: Validation of findings in real-life, high-risk context in credit scoring at a major Belgian bank.

Purpose: To empirically validate the FAIR-XAIN framework on customer data from a major Belgian bank, and to assess how fair XAIN are perceived and used by both providers and affected individuals.

Tasks: **(T3.1) Validation of research findings:** I will go through the whole pipeline as given in Figure 2 based on the data provided by the bank. The finally generated XAIN will be evaluated on FAIR-XAIN fairness scores and debiased with the proposed mitigation techniques. **(T3.2) Stakeholder evaluation study:** To assess practical relevance and usability, a mixed evaluation will be conducted consisting of semi-structured interviews with 5-8 data scientists and credit analysts, and a controlled client study in which participants are randomly assigned to receive either fair XAIN, unmodified XAIN, or raw technical XAI artefacts. Interviews will explore implementability and perceived trade-offs, while the client study will measure clarity, trust, ease-of-use, and decision comprehension using structured questionnaires aligned with prior evaluation formats (Martens, Hinns, et al., 2025).

Deliverables: **(D3.1)** Real-world validation of FAIR-XAIN in credit decisioning. **(D3.2)** Empirical evidence of fair XAIN implementation versus unmodified XAIN and raw XAI artefacts by experts and lay users.

Transparency and reproducibility

To ensure reproducibility and facilitate benchmarking, the FAIR-XAIN framework, including both fairness assessment (WP1) and mitigation practices (WP2), will be released as **open-source on GitHub**, in line with the rest of our research group's work. All datasets on which the framework will be initially developed are publicly available and often used in the fairness fields. In addition to source code, the repository will include standardized experimental pipelines, documented prompt templates, and evaluation protocols, **enabling full replication of reported results**. All model configurations, LLM parameters, and FAIR-XAIN settings will be explicitly logged. This will allow other researchers and practitioners to reproduce findings, compare mitigation strategies, and conduct large-scale benchmarking studies across models and datasets. Where real-world bank data cannot be shared (WP3), equivalent experiments will be replicated on public datasets to preserve transparency.

Risk and mitigation measures

A first risk to account for is the potential **absence of unfairness** (WP1), although first experiments do indeed show clear differences. This will shift the research focus towards the design choices that (don't) give rise to fairness. Additionally, the proposed framework can in any way still detect unfairness in future XAIN and LLM developments, positioning it as a forward-looking auditing framework. The mitigation strategies (WP2) will then be developed and tested based on artificially introduced unfairness. Secondly, the proposed mitigation strategies could show **limited effectiveness**. In this case, combinations of strategies and stricter agentic constraints will be implemented. There could also be difficulties in **recruiting experts** for the qualitative evaluation (WP3), which I would then replace with online panels of data scientists from the bank sector. A final risk is the rapid **pace of development of LLMs** (WP1+WP2). Advances in model (LLM) architectures and capabilities may occur faster than the research cycle, potentially reducing the long-term applicability of specific technical findings. Nevertheless, this research focuses on a decision-making framework rather than on model-specific performance alone. These insights are thus easily transferable across different generations of LLMs.

References

- Alvarez-Melis, D., & Jaakkola, T. S. (2018). [On the Robustness of Interpretability Methods](#). Arxiv preprint.
- Bai, X., Wang, A., Sucholutsky, I., & Griffiths, T. L. (2025). [Explicitly unbiased large language models still form biased associations](#). *Proceedings of the National Academy of Sciences*, 122(8), e2416228122.
- Barocas, S., Hardt, M., & Narayanan, A. (2023). [Fairness and Machine Learning: Limitations and Opportunities](#). MIT Press.
- Bilal, A., Ebert, D., & Lin, B. (2025). [LLMs for Explainable AI: A Comprehensive Survey](#). Arxiv.
- Binns, R. (2020). [On the apparent conflict between individual and group fairness](#). In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (pp. 514–524).
- Bolukbasi, T., Chang, K.-W., Zou, J. Y., Saligrama, V., & Kalai, A. T. (2016). [Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings](#). *Advances in Neural Information Processing Systems*, 29.
- Brughmans, D., Leyman, P., & Martens, D. (2024). [NICE: An algorithm for nearest instance counterfactual explanations](#). *Data Mining and Knowledge Discovery*, 38(5), 2665–2703.
- Caliskan, A., Bryson, J. J., & Narayanan, A. (2017). [Semantics derived automatically from language corpora contain human-like biases](#). *Science*, 356(6334), 183–186.
- Gallegos, I. O., Rossi, R. A., Barrow, J., Tanjim, M. M., Kim, S., Deroncourt, F., Yu, T., Zhang, R., & Ahmed, N. K. (2024). [Bias and fairness in large language models: A survey](#). *Computational Linguistics*, 50(3), 1097–1179.
- Gao, Z., Tong, L., & Zhang, Z. (2026). [Detecting and Evaluating Bias in Large Language Models: Concepts, Methods, and Challenges](#). *Journal of Behavioral Data Science*, 6(1), 1–68.
- Goethals, S., Martens, D., & Evgeniou, T. (2023). [Manipulation Risks in Explainable AI: The Implications of the Disagreement Problem](#). In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases* (pp. 185–200). Cham: Springer Nature Switzerland.
- Ghosh, S., & Wilson, K. (2025). [Bias Is a Math Problem, AI Bias Is a Technical Problem: 10-Year Literature Review of AI/LLM Bias Research Reveals Narrow \[Gender-Centric\] Conceptions of 'Bias', and Academia-](#)

- [Industry Gap](#). In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* (Vol. 8, No. 2, pp. 1091-1106).
- Ichmourkamedov, T., Hinns, J., & Martens, D. (2024). [How good is my story? Towards quantitative metrics for evaluating LLM-generated XAI narratives](#). Arxiv preprint.
- Kavouras, L., Tsopeles, K., Giannopoulos, G., Sacharidis, D., Psaroudaki, E., Theologitis, N., Rontogiannis, D., Fotakis, D., & Emiris, I. (2023). [Fairness aware counterfactuals for subgroups](#). *Advances in Neural Information Processing Systems*, 36, 58246–58276.
- Leidinger, A., & Rogers, R. (2024). [How Are LLMs Mitigating Stereotyping Harms? Learning from Search Engine Studies](#). *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 7(1), 839–854.
- Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). [RoBERTa: A Robustly Optimized BERT Pretraining Approach](#). Arxiv preprint.
- Lo Piano, S. (2020). [Ethical principles in machine learning and artificial intelligence: Cases from the field and possible ways forward](#). *Humanities and Social Sciences Communications*, 7(1), 9.
- Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., & Lee, S.-I. (2020). [From local explanations to global understanding with explainable AI for trees](#). *Nature Machine Intelligence*, 2(1), 56–67.
- Lundberg, S. M., & Lee, S.-I. (2017). [A Unified Approach to Interpreting Model Predictions](#). *Advances in Neural Information Processing Systems*, 30.
- Martens, D., Hinns, J., Dams, C., Vergouwen, M., & Evgeniou, T. (2025). [Tell me a story! Narrative-driven XAI with Large Language Models](#). *Decision Support Systems*, 191, 114402.
- Martens, D., Shmueli, G., Evgeniou, T., Bauer, K., Janiesch, C., Feuerriegel, S., Gabel, S., Goethals, S., Greene, T., Klein, N., Kraus, M., Kühl, N., Perlich, C., Verbeke, W., Zharova, A., Zschech, P., & Provost, F. (2025). [Beware of “Explanations” of AI](#). Arxiv preprint.
- Mill, E., Garn, W., & Turner, C. (2024). [Real-World Efficacy of Explainable Artificial Intelligence using the SAGE Framework and Scenario-Based Design](#). *Applied Artificial Intelligence*, 38(1), 2430867.
- Miller, T., Howe, P., & Sonenberg, L. (2017). [Explainable AI: Beware of Inmates Running the Asylum Or: How I Learnt to Stop Worrying and Love the Social and Behavioural Sciences](#). Arxiv preprint.
- Molnar, C. (2025). [A Guide for Making Black Box Models Explainable \(3rd ed.\)](#).
- Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the [Protection of Natural Persons with Regard to the Processing of Personal Data and on the Free Movement of Such Data](#), and Repealing Directive 95/46/EC (General Data Protection Regulation) (2016). *Official Journal of the European Union*, L 119, 1–88.
- Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 Laying down [Harmonised Rules on Artificial Intelligence and Amending Regulations](#) (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act) (2024).
- Rong, Y., Leemann, T., Nguyen, T.-T., Fiedler, L., Qian, P., Unhelkar, V., Seidel, T., Kasneci, G., & Kasneci, E. (2024). [Towards Human-Centered Explainable AI: A Survey of User Studies for Model Explanations](#). *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(4), 2104–2122.
- Saarela, M., & Podgorelec, V. (2024). [Recent Applications of Explainable AI \(XAI\): A Systematic Literature Review](#). *Applied Sciences*, 14(19).
- Silvestri, M., Vineis, V., Gabrielli, E., Giorgi, F., Veglianti, F., Silvestri, F., & Tolomei, G. (2025). [A Survey on Explainable AI Narratives based on Large Language Models](#). Techrxiv preprint.
- Slack, D., Hilgard, S., Singh, S., & Lakkaraju, H. (2021). [Reliable Post hoc Explanations: Modeling Uncertainty in Explainability](#). *Advances in Neural Information Processing Systems*, 34, 9391-9404.
- Suresh, H., & Gutttag, J. (2021). [A Framework for Understanding Sources of Harm throughout the Machine Learning Life Cycle](#). *Proceedings of the 1st ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*, EAAMO '21, 1–9.
- Wachter, S., Mittelstadt, B., & Russell, C. (2017). [Counterfactual Explanations without Opening the Black Box: Automated Decisions and the GDPR](#). *Harvard Journal of Law & Technology (Harvard JOLT)*, 31(2), 841–888.
- Zhao, H., Chen, H., Yang, F., Liu, N., Deng, H., Cai, H., Wang, S., Yin, D., & Du, M. (2023). [Explainability for Large Language Models: A Survey](#). *ACM Transactions on Intelligent Systems and Technology*, 15(2), 1–38.